

Lab 07.01: User and Group Accounts (part I)

Practice Contents

Start your new VM, open a terminal window and log in to your account. **su** to root (using your new root password).

```
trietnm@OSG203:/home/trietnm
[trietnm@OSG203 ~]$ su
Password:
[root@OSG203 trietnm]#
```

1. Using the Cockpit GUI. We have to make sure Cockpit is installed and running.

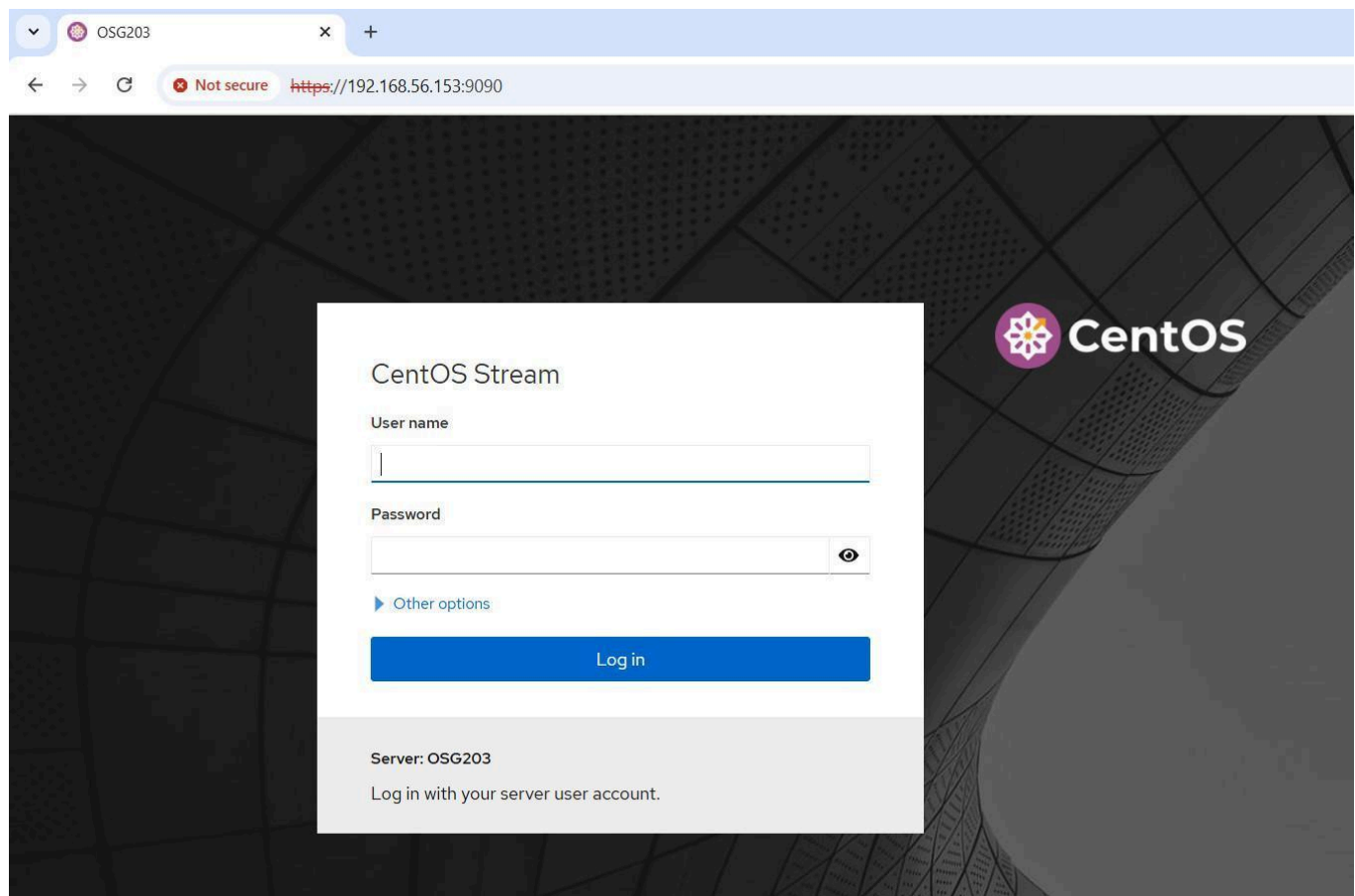
- a. In your terminal window, type **whereis cockpit**. You should find it in `/usr/share/cockpit`.

```
[root@OSG203 trietnm]# whereis cockpit
cockpit: /etc/cockpit /usr/share/cockpit
```

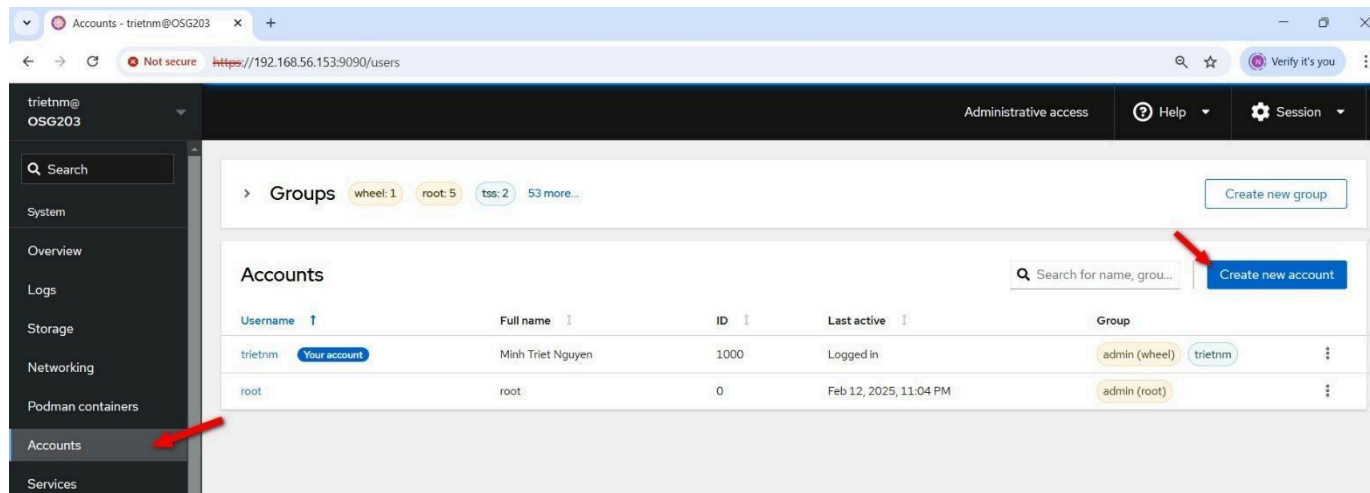
If not, type **dnf -y install cockpit**. Next, type **systemctl status cockpit.socket** to see if it is loaded, active and listening. If not, type **systemctl enable cockpit.socket** followed by **systemctl start cockpit.service**. Next, type **firewall-cmd --info-zone=public** and see if cockpit is listed under services. If not, type **firewall-cmd --add-service=cockpit --permanent** followed by **firewall-cmd --reload**. Make sure your network service is running (you can determine this by opening the Firefox web browser and seeing if you can connect to any page). If it is not, under **Activities** select the **Apps** selection and from there, select **Settings**. From the Settings window, select **Network** and **enable** it.

- b. Type **ip addr** to obtain your IPv4 address. This will be listed under `inet` probably as the second or third device, possibly named `ens32`, `enp0s3`,... or some variation (the first device is denoted as `lo`). Now, open your web browser and enter the URL **`https://ipaddress:9090`** where *ipaddress* is the IPv4 address you found from `ip addr`. You will be told this is a security risk because you are using `https` and your computer does not have an X.509 certificate. **Accept** the risk by selecting **Advanced** and further down on the page, **Accept the Risk** and **Continue**. Once connected to the Cockpit interface, you will be asked to log in.

```
[root@OSG203 trietnm]# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
   inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
   inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
   link/ether 08:00:27:99:7f:5f brd ff:ff:ff:ff:ff:ff
   inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
       valid_lft 85549sec preferred_lft 85549sec
   inet6 fd00::a00:27ff:fe99:7f5f/64 scope global dynamic noprefixroute
       valid_lft 86073sec preferred_lft 14073sec
   inet6 fe80::a00:27ff:fe99:7f5f/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
   link/ether 08:00:27:2c:af:d2 brd ff:ff:ff:ff:ff:ff
   inet 192.168.56.153/24 brd 192.168.56.255 scope global dynamic noprefixroute enp0s8
       valid_lft 349sec preferred_lft 349sec
   inet6 fe80::a00:27ff:fe2c:afd2/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
```

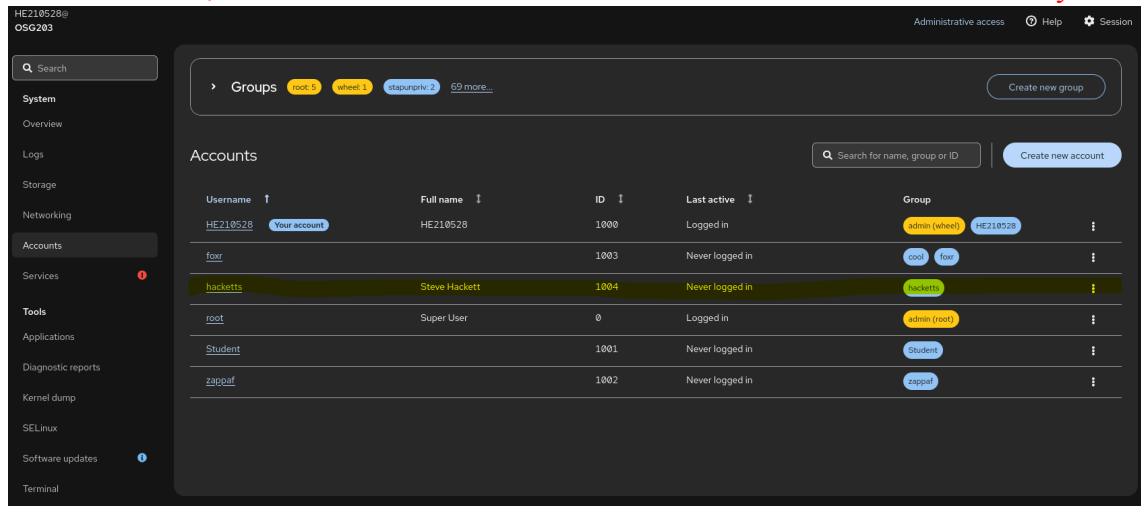


Login as your account. Select **Log In**. In the menu on the left, select **Accounts**. You will see two accounts, root and the account you created. Select **Create New Account**.



From the Create New Account window, create an account for the new user Steve Hackett. Change the username from shackett to **hacketts**. Provide the password **guitar** and confirm it. Select **Create**. *What happens?*

Sẽ có 1 thông báo mà mật khẩu quá ngắn và hiện lên 1 nút “tạo tài khoản với mật khẩu yếu” màu vàng và sau khi ấn chọn thì sẽ có 1 tài khoản có username à hacketts và tên đầy đủ là Steve Hackett



Create new account

Full name Steve Hackett

User name shacketts

Home directory /home/shacketts

Shell /bin/bash

User ID 1001

Authentication
☒ Use password
☐ Require password change on first login
☐ Disallow password authentication ?

Password guitar
Weak password

Confirm password guitar

Create **Cancel**

You can't use such a short password. So try **guitarist** for the password, **confirm** it and select **Create**.
What happens this time?

Vẫn sẽ hiện lên là mật khẩu yếu nhưng vẫn cho phép tạo

The screenshot shows a 'Create new account' form with the following fields and values:

- User name: hacketts
- Home directory: /home/hacketts
- Shell: /bin/bash
- User ID: 1005
- Authentication:
 - ☒ Use password
 - ☐ Require password change on first login
 - ☐ Disallow password authentication ⓘ
- Password: guitarist (with a 'Weak password' indicator below the field)
- Confirm password: guitarist

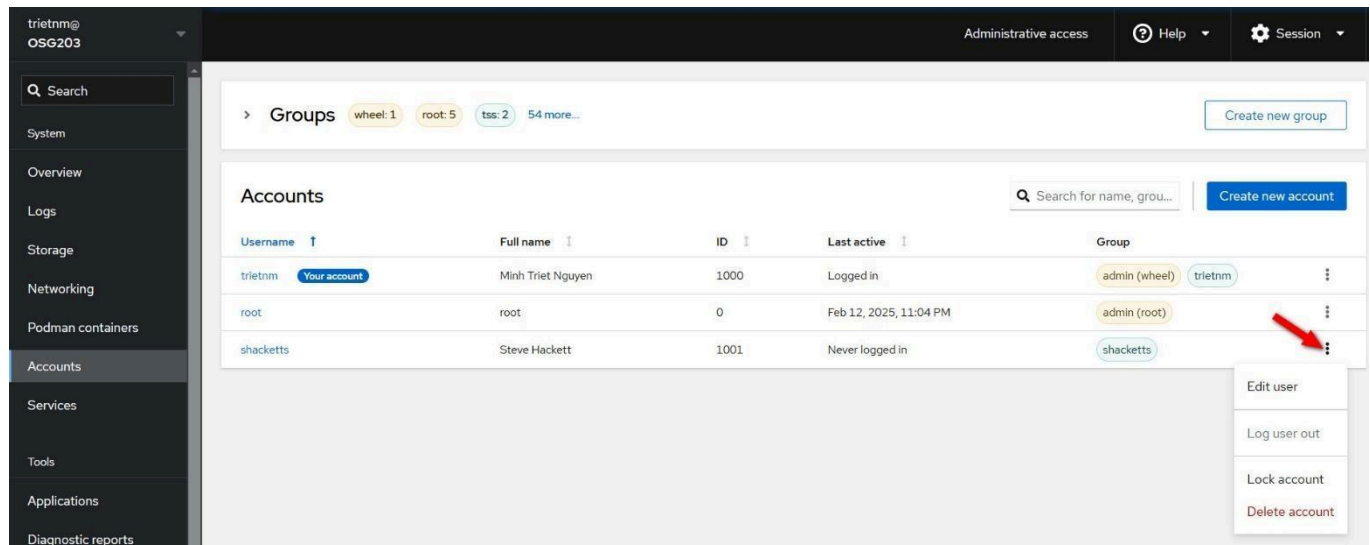
At the bottom, there are three buttons: 'Create' (disabled), 'Create account with weak password' (highlighted in yellow), and 'Cancel'.

Delete what you had and try to create the account with no password. *What happens?*

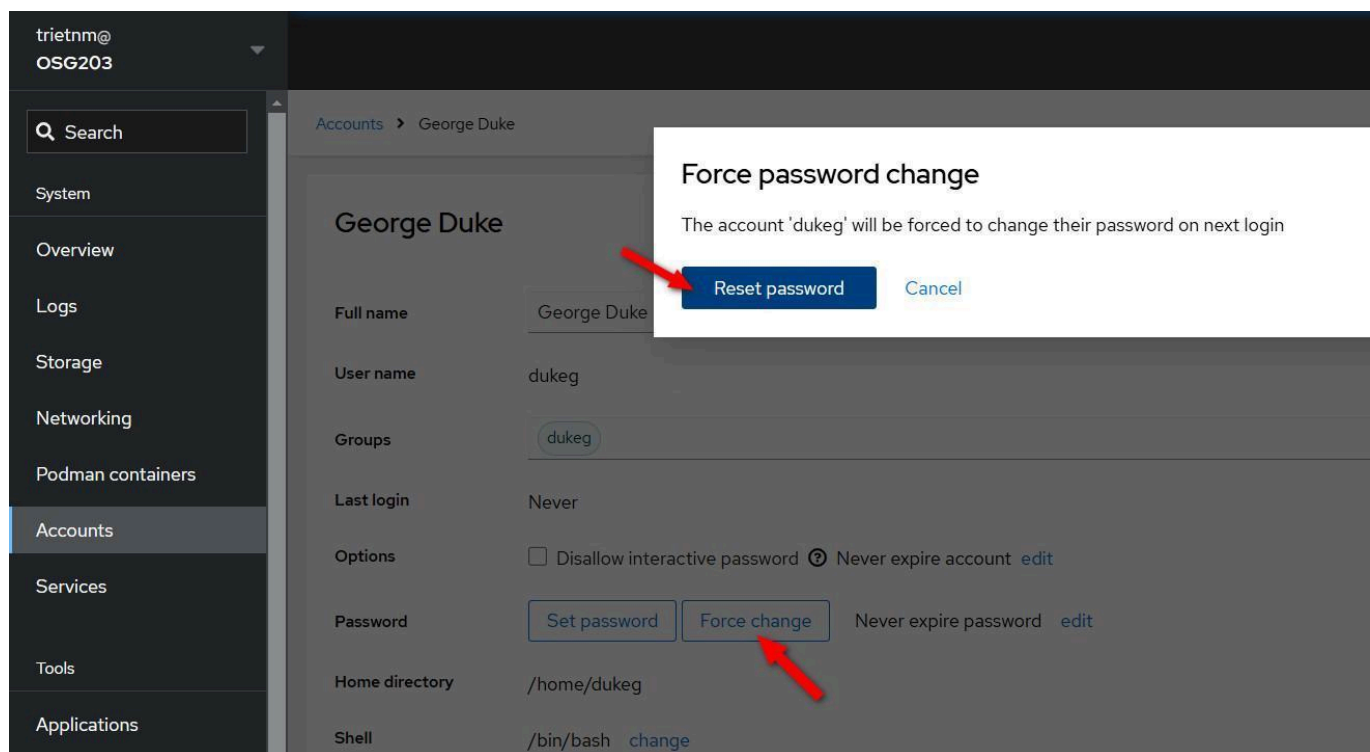
Sẽ có thông báo là mật khẩu trống và nút vàng vẫn ở đây nhưng khi ấn vào thì không thay đổi gì cả

The screenshot shows the same 'Create new account' form, but with the password field empty. A yellow warning triangle icon is visible next to the password field, and the text 'Empty password' is displayed below it. The 'Confirm password' field is also empty. The 'Create account with weak password' button remains highlighted in yellow.

Having created the account, we are returned to the list of users. Select his account from the list. From here, we can delete the account, terminate his current session (if he were logged in, which he isn't), assign him to an administrator role, lock his account, set his password or select Force Change, which requires him to change password when he first logs in. We can also specify that his account will never be locked and his password will never expire.

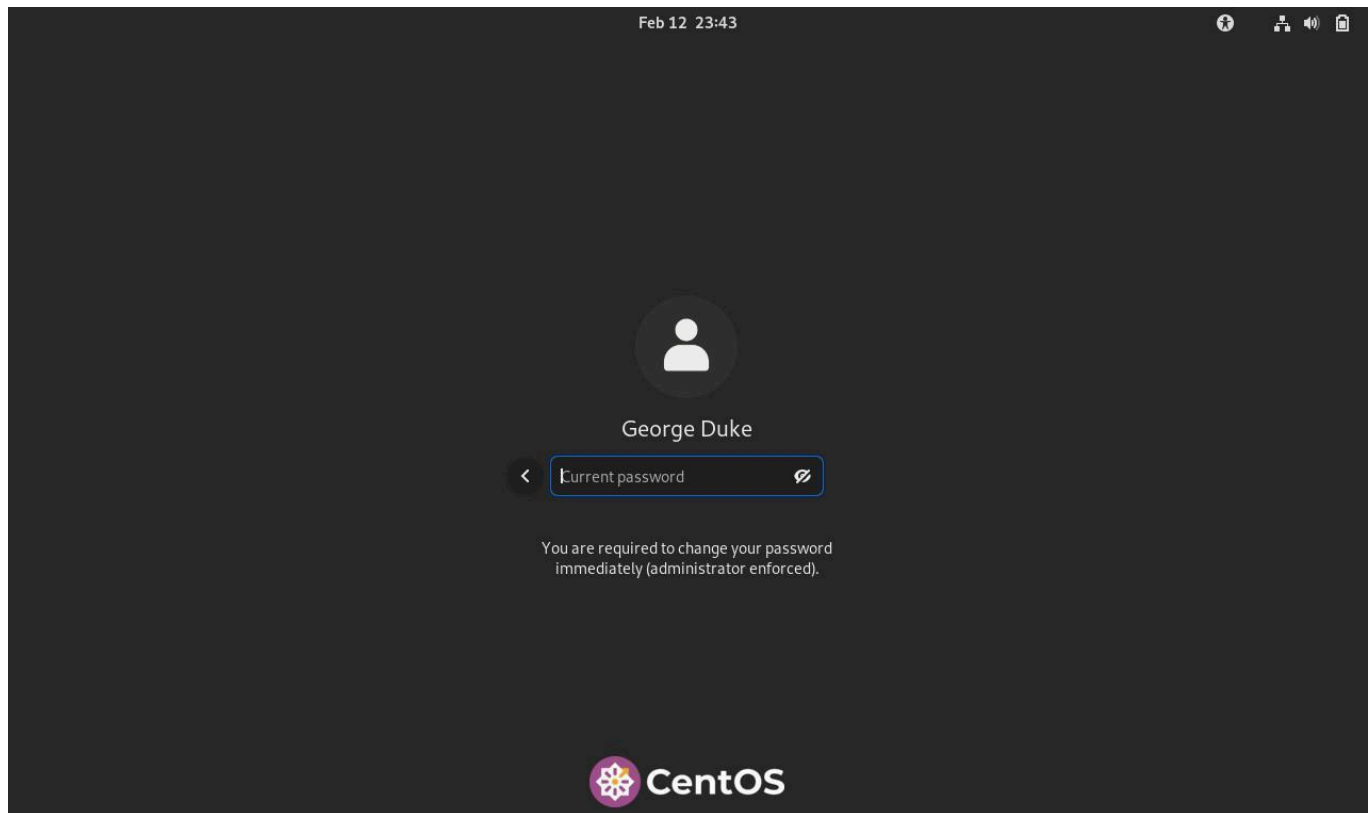


- c. At the top of the window, select Accounts to return to the main accounts window. Here, create a new account for George Duke, changing the username to dukeg, assign a password of Eaou&Duewtt1 and Create. Having been returned to the list of users, select this account and select Force Change. A popup window asks you to confirm this by selecting Reset. Do so.



We will now log in as dukeg to see that we are forced to change the password. From the upper right, select the down arrow icon and from the list of choices, select your account name. From the pop-up list, select Switch User. You will be shown a list of users, select George Duke and log in (Eaou&Duewtt1) and you will be asked to change password. To change the password, you are asked again for the current password and then the new password. Use FZmoi70! (and confirm it). You are logged in as George Duke and shown the Welcome screen as this is this user's first time logging in. Step through these screens and when it

closes, you are shown the Gnome Help screen, close it. Log out and log back in as yourself. Open a terminal window and su to root.



2. Create users and groups through the command line using useradd and groupadd.

Before beginning, we will install two shells that may not be pre-installed. Type **which csh** and **which tcsh**, if the response is no tcsh/csh in (...) for either or both, then install them using type **dnf -y install tcsh** (this installs one or both if one is missing).

```
[root@OSG203 trietnm]# which csh
/usr/bin/which: no csh in (/root/.local/bin:/root/bin:/home/trietnm/.local/bin:/home/trietnm/bin:/usr/local/bin:/usr/bin:/usr/local/sbin:/usr/sbin)
[root@OSG203 trietnm]# which tcsh
/usr/bin/which: no tcsh in (/root/.local/bin:/root/bin:/home/trietnm/.local/bin:/home/trietnm/bin:/usr/local/bin:/usr/bin:/usr/local/sbin:/usr/sbin)
```

a. Type **useradd**. This displays the instruction options. It requires at least the username. Add **-m** to create a home directory. *Under what circumstance might you not want to create a home directory for a user?* Other useful useradd options are **-c**, **-d**, **-e**, **-G**, and **-s**. *What do each of these options do?*

- **-c**: Thêm thông tin mô tả về người dùng (thường là tên đầy đủ) vào trường GECOS trong tệp `/etc/passwd`.
- **-d**: Chỉ định một đường dẫn cụ thể làm thư mục home cho người dùng thay vì sử dụng đường dẫn mặc định (`/home/username`).
- **-e**: Đặt ngày hết hạn cho tài khoản (định dạng YYYY-MM-DD). Sau ngày này tài khoản sẽ bị khóa.
- **-G**: Thêm người dùng vào danh sách các nhóm phụ (Supplementary groups).
- **-s**: Chỉ định login shell mặc định cho người dùng khi đăng nhập.

*Under what circumstance(s) might you want to use the option **-M**?*

Khi muốn tạo một tài khoản người dùng thông thường nhưng cố tình không muốn hệ thống tự động tạo thư mục home cho tài khoản đó.

*Under what circumstance(s) might you want to use the option **-u**?*

Khi cần gán một số định danh User ID (UID) cụ thể cho người dùng

Under what circumstance(s) might you want to use the options `-u -o`?

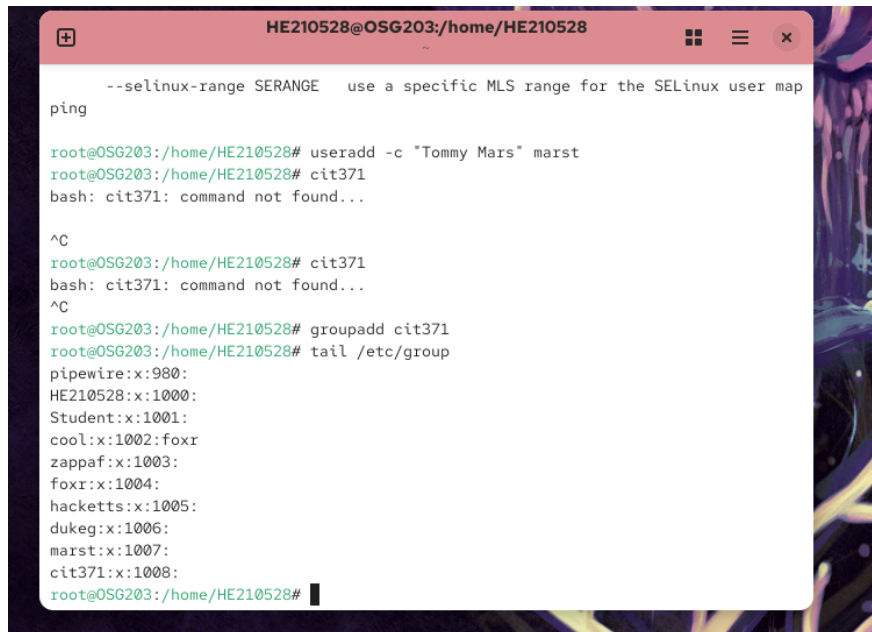
Khi muốn tạo một tài khoản người dùng có UID trùng với một UID đã tồn tại sẵn trên hệ thống (chủ yếu dùng để tạo ra một tài khoản phụ có chung quyền hạn tối cao với tài khoản root - UID 0)

- b. Create an account for Tommy Mars (marst), inserting his full name as a comment in the GECOS (comment) field. Don't forget that the name has to be put in quote marks since it has a blank space. *What command did you enter?*

`useradd -c "Tommy Mars" marst`

- c. Next, we create a group with the command **groupadd cit371**. This creates a group using the next available group ID (GID). Type **tail /etc/group** to see what ID was given to this group. *What is its GID?*

`cit371:x:1008:`



```
HE210528@OSG203:/home/HE210528
--selinux-range SERANGE use a specific MLS range for the SELinux user map
ping
root@OSG203:/home/HE210528# useradd -c "Tommy Mars" marst
root@OSG203:/home/HE210528# cit371
bash: cit371: command not found...

^C
root@OSG203:/home/HE210528# cit371
bash: cit371: command not found...

^C
root@OSG203:/home/HE210528# groupadd cit371
root@OSG203:/home/HE210528# tail /etc/group
pipewire:x:980:
HE210528:x:1000:
Student:x:1001:
cool:x:1002:foxr
zappaf:x:1003:
foxr:x:1004:
hacketts:x:1005:
dukeg:x:1006:
marst:x:1007:
cit371:x:1008:
root@OSG203:/home/HE210528#
```

Now, create an account for Ruth Underwood of underwoodr, giving her name in the GECOS field and adding her to the cit371 group. *What command did you enter?*

`useradd -c "Ruth Underwood" -G cit371 underwoodr`

Type **tail /etc/passwd** and look at the two numbers that follow `:x:`, these are this user's UID and GID. *Why are they different?*

Vì khi chạy lệnh trên, hệ thống tự động tạo ra một Nhóm User Private Group có tên là `underwoodr` và gán GID của nhóm chính này cho cô ấy (trùng với UID). Còn nhóm `cit371` mà chỉ định bằng tùy chọn `-G` chỉ đóng vai trò là một nhóm phụ (Supplementary group) nên thông số GID của nhóm phụ này sẽ không xuất hiện tại vị trí số thứ hai trong `/etc/passwd`

Type **passwd underwoodr** to assign the account an initial password. Use the password **xylophone**. *What warning message did you get because of the password?*

BAD PASSWORD: The password fails the dictionary check - it is based on a dictionary word

```
underwood1.x.1007.1009.ruth underwood:/home/underwood1:/bin/bash
root@OSG203:/home/HE210528# passwd underwoodr
New password:
BAD PASSWORD: The password fails the dictionary check - it is based on a dictionary word
Retype new password:
```

- d. Use groupadd to create two groups: students and dummies. For dummies, give it a GID of 2000. *What commands did you enter?*

groupadd students

groupadd -g 2000 dummies

- e. Create the following users with the given names and usernames and use the defaults for everything else except as noted. *Place the instruction used into your answers.*

- i. Suzie Creamcheese, creamcheeses (username), default login shell /usr/bin/tcsh.

useradd -c "Suzie Creamcheese" -s /usr/bin/tcsh creamcheeses

- ii. Eric Cartman, cartmane (username), groups cit371 and students, UID of 1501.

useradd -c "Eric Cartman" -G cit371,students -u 1501 cartmane

- iii. Kate Bush, bushk (username), set her expiration date to January 1, 2028 (the date is specified as 2028-01-01).

useradd -c "Kate Bush" -e 2028-01-01 bushk

- iv. CIT 371 Student, cit371 (username), no directory. *What error did this give you?*

Lỗi: useradd: group cit371 exists - if you want to add this user to that group

To fix the problem, create the account with no private group.

useradd -c "CIT 371 Student" -M -N -g cit371 cit371

- v. Use passwd (4 times) to give these four users initial passwords of xyz12abc, cheesypoofs, 9thWave!, and a password of your own making respectively. *What warnings did each give you if any?*

- **User creamcheeses:**

- + **Cảnh báo: BAD PASSWORD: The password fails the dictionary check - it is too simplistic/systematic**

- **User cartmane:**

- + **Cảnh báo: BAD PASSWORD: The password fails the dictionary check - it is based on a dictionary word**

- **User bushk và cit371:**

+Cảnh báo: Không có cảnh báo nào xuất hiện

```

root@OSG203:/home/HE210528# passwd creamcheeses
New password:
BAD PASSWORD: The password fails the dictionary check - it is too simplistic/syst
ematic
Retype new password:
passwd: password updated successfully
root@OSG203:/home/HE210528# passwd cartmane
New password:
BAD PASSWORD: The password fails the dictionary check - it is based on a dictiona
ry word
Retype new password:
passwd: password updated successfully
root@OSG203:/home/HE210528# passwd bushk
New password:
Retype new password:
passwd: password updated successfully
root@OSG203:/home/HE210528# passwd cit371
New password:
Retype new password:
passwd: password updated successfully
root@OSG203:/home/HE210528#

```

3. Explore how to create a larger number of accounts through a shell script.

- a. Write the shell script from chapter 7 shown in figures 7.6, 7.7 and 7.10 replacing the `apg` command from figure 7.10 with `date +%s | sha256sum | head -c10`. The reason for this is that `apg`, the command used in the textbook to create passwords, is not installed on your computer. Type the above `date` command from the command line to see what it does. *Explain this command* (you might need to read the man pages for `date` and `sha256sum`). Save your script and change its permissions so it is executable.

Example:

- Create a new script file using the command `nano create_users.sh` Then paste the following content into the script file and save it.

```

#!/bin/bash

# Loop to read each line for first name and last name
while read first last; do
    # Combining first and last name as the full name
    name="$first $last"

    # Creating username from the first letter of first name and the full last name
    username="{last}${first:0:1}"

    # Check if the username already exists in the system
    n=$(grep -c "^$username:" /etc/passwd)

    # If username exists, append the count to make it unique
    if [ $n -gt 0 ]; then
        username="{username}${(n+1)}"
    fi

    # Adding the user with the specified full name
    useradd -c "$name" -m $username

```

```
# Generating a password using the date and sha256sum commands
password=$(date +%s | sha256sum | head -c10)
```

```
# Setting the password for the user
echo "$username:$password" | chpasswd
```

```
# Echo the username and password to a file for record-keeping
echo "$username $password" >> /root/tempPasswords
```

```
done < $1 # Reading from a file passed as an argument to the script
```

- Grant execute permissions to the script file using the command **chmod +x create_users.sh**

- b. Type **wget www.nku.edu/~foxr/accounts.txt** to download this textfile of names. Run your script, redirecting accounts.txt as input (using **./create_users.sh accounts.txt**). This should generate new accounts for these. You should receive output about each new user's password being updated. Hopefully you will not get error messages. If you do get errors, see if any of the accounts were created. If so, delete the file **/root/tempPasswords** and delete all of the new user accounts (by hand using **userdel -r username**). Try to figure out what went wrong in your script, fix it and rerun the script. Repeat this until the script runs successfully, creating the 14 new accounts. Once the accounts have been created, type **tail -n 23 /etc/passwd** and *place the resulting output in your answers* as the answer to this step. Also type **cat /root/tempPasswords** and place the resulting output in your answers (this is the list of passwords created for the new users).

4. Understanding User Home Directories and Permissions in Linux.

- Unless specified otherwise, all new users are given their own home directories. Type **ls -l /home**. *Which of the users created in this lab does not have a home directory and why?*

Người dùng **cit371** không có thư mục home, bởi vì tài khoản này được tạo bằng tùy chọn không tạo thư mục riêng

You will find every directory is identical in its size and creation date except for yours. How are these directories created?

Khi một tài khoản mới được tạo có yêu cầu thư mục home, hệ thống sẽ tự động sao chép toàn bộ cấu trúc cây thư mục ẩn và các tệp cấu hình mẫu nằm trong thư mục gốc **/etc/skel** sang thư mục đích **/home/username**

- When a new user account is generated, if it is given its own home directory, it is created by copying the files from **/etc/skel**. Type **ls -al /etc/skel** and **ls -al /home/user** for any *user* found in **/home** (other than yours). *What are the items found in /etc/skel and in that user's directory?*

Gồm các tệp ẩn khởi tạo môi trường shell cho người dùng như: **.bash_logout**, **.bash_profile**, **.bashrc**. Một số hệ thống có thêm thư mục **.mozilla** để lưu cấu hình trình duyệt

One item is a directory, what is it and what does it store? In comparing the contents of **/etc/skel** to the user you selected, there is one difference (not in the files but in the file permissions). *What is that difference?*

Các tệp trong **/etc/skel** thuộc sở hữu của **root:root** và có quyền đọc chung cho mọi người. Khi được sao chép sang thư mục của user, hệ thống tự động đổi quyền sở hữu (Ownership) về cho chính **username:username**, đồng thời thắt chặt phân quyền (Permissions) của thư mục home (thường là **700** hoặc **750**) để các người dùng khác không thể tùy ý đọc dữ liệu bên trong

(ignore the number of hard links for the parent directory, **..**). Also created automatically for each user is a mail "spool" file, found in **/var/spool/mail**. Type **ls -l** on this directory. *What do you see here? What are the sizes of these entries?* One item is a mailbox of a user that you did not create in this lab. *Which item?*

Sẽ thấy các tệp rỗng (kích thước 0 bytes) có tên trùng khớp với từng username trên hệ thống. Riêng tệp mailbox của người dùng đặc biệt không tạo trong lab này thường là **rpc** hoặc **chrony**

Notice that each of these mailbox files is owned by the program mail but each has its own user name for the group. *Why do you suppose these files have a different group owner?*

Để đảm bảo tính riêng tư và bảo mật. Group Owner của tệp mailbox được gán chính là nhóm riêng của người dùng đó (User Private Group). Cơ chế này giúp tiến trình mail của hệ thống (chạy bằng quyền group **mail**) có thể ghi dữ liệu thư vào tệp, trong khi người dùng sở hữu hòm thư vẫn giữ toàn quyền đọc/xóa thư cá nhân mà không bị các tài khoản thường khác đọc trộm

Type **exit** to leave root, close your terminal window and shut down your VM.